Chapter 4: Integers and Rationals

Analysis I, by Terrence Tao

A: The Integers

Definition 1 - Integers ($\mathbb{Z}$)

An integer is an expression of the form $a \text{——} b$, where a and b are natural numbers. Two integers are considered equal ($a \text{——} b = c \text{——} d$) if and only if $a + d = c + b$.

Problem 1 - Equality on the integers follow the three axioms

Verify that the definition of equality on the integers is reflexive, symmetric, and transitive.

Proof. First, we show that the equality is reflexive. Consider the integer $a \text{——} b$. Notice that $a + b = a + b$, so $a \text{——} b = a \text{——} b$.

Second, we prove that the equality is symmetric. Suppose $a \text{——} b = a' \text{——} b'$. Then, $a + b' = a' + b$. Because the equality on the natural numbers is transitive, we have that $a' + b = a + b'$. By definition of the equality on the integers, we have that $a' \text{——} b' = a \text{——} b$.

Next, we show that the equality on the integers is symmetric. Suppose $a \text{——} b = c \text{——} d$ and $c \text{——} d = e \text{——} f$. Then, $a + d = c + b$ and $c + f = e + d$. We add the second equation into the first to get:

$$a + d + c + f = c + b + e + d.$$

By the cancellation law of natural numbers for addition, we can cancel out c and d to get:

$$a + f = b + e.$$

By the definition of equality on the integers, $a \text{——} b = e \text{——} f$. □

Problem 2 - Negation on the Integers is well-defined

Show the definition of negation on the integers is well-defined in the sense that if $(a \text{——} b) = (a' \text{——} b')$, then $-(a \text{——} b) = -(a' \text{——} b')$. Hence, equal integfers have equal negations.

Proof. Suppose $(a \text{——} b) = (a' \text{——} b')$, so $a + b' = a' + b$. Because addition of the natural numbers is commutative, we have that $b' + a = b + a'$. Then, by the symmetricality of the equality of the natural numbers, we have $b + a' = b' + a$. Then, by the definition of the equality on the integers, we have $b \text{——} a = b' \text{——} a'$. Because $b \text{——} a = -(a \text{——} b)$ and $b' \text{——} a' = -(a' \text{——} b')$, we get:

$$-(a \text{——} b) = -(a' \text{——} b').$$

□

Problem 3 - Multiplying by -1 is the same as negation

Show that $(-1) \times a = -a$ for every integer a.

Proof. Recall that $-1 = 0 \text{——} 1$ and $a = (a \text{——} 0)$. Notice:

$$\begin{aligned}
(-1) \times a &= (0 \text{——} 1) \times (a \text{——} 0) \\
&= (0 \times a + 1 \times 0) \text{——} (0 \times 0 + 1 \times a) \\
&= (0 + 0) \text{——} (0 + a) \\
&= 0 \text{——} a \\
&= -a
\end{aligned}$$

□

Problem 4 - Properties of the Integers

Prove the remaining identities in Proposition 4.1.6. For all of the laws of algebra for the integers, let $x = x_1 \mathbin{\text{---}} x_2, y = y_1 \mathbin{\text{---}} y_2, z = z_1 \mathbin{\text{---}} z_2$ be integers, where $x_1, x_2, y_1, y_2, z_1, z_2 \in \mathbb{N}$.

Proposition 4.1. $x + y = y + x$

Proof. Notice:

$$\begin{aligned} x + y &= (x_1 \mathbin{\text{---}} x_2) + (y_1 \mathbin{\text{---}} y_2) \\ &= (x_1 + y_1) \mathbin{\text{---}} (x_2 + y_2) \\ &= (y_1 + x_1) \mathbin{\text{---}} (y_2 + x_2) \\ &= y + x. \end{aligned}$$

$\square$

Proposition 4.2. $(x + y) + z = x + (y + z)$

Proof. Notice:

$$\begin{aligned} (x + y) + z &= \big((x_1 \mathbin{\text{---}} x_2) + (y_1 \mathbin{\text{---}} y_2)\big) \\ &\quad + (z_1 \mathbin{\text{---}} z_2) \\ &= \big((x_1 + y_1) \mathbin{\text{---}} (x_2 + y_2)\big) \\ &\quad + (z_1 \mathbin{\text{---}} z_2) \\ &= \big(x_1 + y_1 + z_1\big) \mathbin{\text{---}} \big(x_2 + y_2 + z_2\big) \\ &= \big(x_1 + (y_1 + z_1)\big) \\ &\quad \mathbin{\text{---}} \big(x_2 + (y_2 + z_2)\big) \\ &= \big(x_1 \mathbin{\text{---}} x_2\big) \\ &\quad + \big((y_1 + z_1) \mathbin{\text{---}} (y_2 + z_2)\big) \\ &= x + (y + z). \end{aligned}$$

$\square$

Proposition 4.3. $x + 0 = 0 + x = x$

Proof. Recall that $x = (x \mathbin{\text{---}} 0)$ and $0 = (0 \mathbin{\text{---}} 0)$. Notice:

$$\begin{aligned} x + 0 &= (x \mathbin{\text{---}} 0) + (0 \mathbin{\text{---}} 0) \\ &= (x + 0) \mathbin{\text{---}} (0 + 0) \\ &= x \mathbin{\text{---}} 0 \\ &= x. \end{aligned}$$

Similarly, because addition is commutative, we have that $0 + x = x$. $\square$

Proposition 4.4. $x + (-x) = (-x) + x = 0$

Proof. Notice:

$$\begin{aligned} x + (-x) &= (x_1 \mathbin{\text{---}} x_2) \\ &\quad + (x_2 \mathbin{\text{---}} x_1) \\ &= (x_1 + x_2) \mathbin{\text{---}} (x_2 + x_1) \\ &= (x_1 + x_2) \mathbin{\text{---}} (x_1 + x_2) \\ &= 0. \end{aligned}$$

Because addition is commutative, we have that $(-x) + x = 0$. $\square$

Proposition 4.5. $xy = yx$

Proof. Notice:

$$\begin{aligned} xy &= (x_1 \mathbin{\text{---}} x_2) \times (y_1 \mathbin{\text{---}} y_2) \\ &= (x_1y_1 + x_2y_2) \mathbin{\text{---}} (x_1y_2 + x_2y_1) \\ &= (y_1x_1 + y_2x_2) \mathbin{\text{---}} (y_2x_1 + y_1x_2) \\ &= (y_1x_1 + y_2x_2) \mathbin{\text{---}} (y_1x_2 + y_2x_1) \\ &= yx. \end{aligned}$$

$\square$

Proposition 4.6. $(xy)z = x(yz)$.

Proof. This has already been done in the book. $\square$

Proposition 4.7. $x1 = 1x = x$

Proof. Recall that $1 = (1 \mathbin{\text{---}} 0)$. Notice:

$$\begin{aligned} 1x &= (1 \mathbin{\text{---}} 0) \times (x_1 \mathbin{\text{---}} x_2) \\ &= (1x_1 + 0x_2) \mathbin{\text{---}} (1x_2 + 0x_1) \\ &= x_1 \mathbin{\text{---}} x_2 \\ &= x. \end{aligned}$$

Bcause multiplication is commutative, $x1 = x$. $\square$

Proposition 4.8. $x(y + z) = xy + xz$

Proof. Notice:

$$\begin{aligned} x(y + z) &= \big(x_1 \mathbin{\text{---}} x_2\big)\big((y_1 \mathbin{\text{---}} y_2) + (z_1 \mathbin{\text{---}} z_2)\big) \\ &= \big(x_1 \mathbin{\text{---}} x_2\big)\big((y_1 + z_1) \mathbin{\text{---}} (y_2 + z_2)\big) \\ &= \big(x_1(y_1 + z_1) + x_2(y_2 + z_2)\big) \\ &\quad \mathbin{\text{---}} \big(x_1(y_2 + z_2) + x_2(y_1 + z_1)\big) \\ &= \big(x_1y_1 + x_1z_1 + x_2y_2 + x_2z_2\big) \\ &\quad \mathbin{\text{---}} \big(x_1y_2 + x_1z_2 + x_2y_1 + x_2z_1\big) \\ &= \big((x_1y_1 + x_2y_2) + (x_1z_1 + x_2z_2)\big) \\ &\quad \mathbin{\text{---}} \big((x_1y_2 + x_2y_1) + (x_1z_2 + x_2z_1)\big) \\ &= \big((x_1y_1 + x_2y_2) \mathbin{\text{---}} (x_1y_2 + x_2y_1)\big) \\ &\quad + \big((x_1z_1 + x_2z_2) \mathbin{\text{---}} (x_1z_2 + x_2z_1)\big) \\ &= xy + xz \end{aligned}$$

$\square$

Proposition 4.9. $(y + z)x = yx + zx$

Proof. Notice:

$$\begin{aligned} (y + z)x &= x(y + z) \\ &= xy + xz \\ &= yx + zx. \end{aligned}$$

$\square$

Problem 5 - Integers have no zero divisors

Let $a, b \in \mathbb{Z}$ Suppose $ab = 0$. Then, $a = 0$ or $b = 0$.

Proof. Suppose that $ab = 0$. We wil use the trichotomy of the integers on the variable a and the cancellation law of the naturals. We wil prove the proposition for all three cases of $a = 0$, a is a positive integer, and a is a negative integer. In all these cases, b is held fixed with $b = b_1 \text{---} b_2$, where $b_1, b_2 \in \mathbb{N}$.

1. Say $a = 0$. Then, the proposition is trivially true.

2. Say a is a positive number, so a takes on the form $a = k \text{---} 0$ for some positive natural number k. Then, $(k \text{---} 0) \times (b_1 \text{---} b_2) = (kb_1 + 0b_2) \text{---} (kb_2 + 0b_1) = kb_1 \text{---} kb_2$. Because $ab = 0$, we have that $(kb_1 \text{---} kb_2) = (0 \text{---} 0)$, which means that $kb_1 = kb_2$. By the cancellation law of the naturals, we have that $b_1 = b_2$, so $b = 0$.

3. Say a is a negative number, so a takes on the form $a = 0 \text{---} k$. Then, $(0 \text{---} k) \times (b_1 \text{---} b_2) = (0b_1 + kb_2) \text{---} (0b_2 + kb_1) = kb_2 \text{---} kb_1$. Because $ab = 0$, we have $(kb_2 \text{---} kb_1) = 0$, so $kb_2 = kb_1$. By the cancellation law, we have that $b_2 = b_1$, so $b = 0$.

□

Problem 6 - Cancellation law for integers

Let $a, b, c \in \mathbb{Z}$ with $c \neq 0$. If $ac = bc$, then $a = b$.

There are two ways to prove this. The first proof uses Prop 4.1.8 where it states that integers have no zero divisors.

Proof. Suppose $ac = bc$. Notice:

$$\begin{aligned}
ac - bc &= 0 && \text{by subtracting } bc \text{ to both sides} \\
c(a - b) &= 0 && \text{by distributing } c \\
a - b &= 0 && \text{by Prop. 4.1.8 since } c \neq 0 \\
a &= b. && \text{by adding } b \text{ to both sides}
\end{aligned}$$

□

The seconed proof uses the trichotomy of the integers on the variable c, and the cancellation law of multiplication for the natural numbers to eventually remove k in the equations.

Proof. We fix $a = a_1 \text{---} a_2$ and $b = b_1 \text{---} b_2$. Since $c \neq 0$, we only consider two cases.

1. Say c is a positive number, so there exists a positive natural number k such that $c = k \text{---} 0$. Notice:

$$\begin{aligned}
ac = bc &\implies (a_1 \text{---} a_2)(k \text{---} 0) = (b_1 \text{---} b_2)(k \text{---} 0) \\
&\implies ka_1 \text{---} ka_2 = kb_1 \text{---} kb_2 \\
&\implies ka_1 + kb_2 = kb_1 + ka_1 \\
&\implies k(a_1 + b_2) = k(b_1 + a_2) \\
&\implies a_1 + b_2 = b_1 + a_2 \\
&\implies a_1 \text{---} a_2 = b_1 \text{---} b_2 \\
&\implies a = b.
\end{aligned}$$

2. Say c is a negative number, so c is the negation $-k$ of a positive natural number k. In other words, $c = 0 \text{---} k$. We get a very similar proof to the second case of c being a positive number. Notice:

$$\begin{aligned}
ac = bc &\implies (a_1 \text{---} a_2)(0 \text{---} k) = (b_1 \text{---} b_2)(0 \text{---} k) \\
&\implies ka_2 \text{---} ka_1 = kb_2 \text{---} kb_1 \\
&\implies k(a_2 + b_1) = k(b_2 + a_1) \\
&\implies a_2 + b_1 = b_2 + a_1 \\
&\implies a_1 \text{---} a_2 = b_1 \text{---} b_2 \\
&\implies a = b.
\end{aligned}$$

□

Lemma 1. *Let $a, b, k \in \mathbb{Z}$. If $a = b + k$, THEN $a = b$ if and only if $k = 0$.*

Proof. Suppose $a = b + k$. Let us prove the forward direction. Assume $a = b$. Then $a = b = b + k$. We subtract b to all sides to get $0 = k$. To prove the converse, assume that $k = 0$. Trivially, $a = b + k = b + 0 = b$. □

Problem 7 - Properties of order for the integers

Prove the properties of order for the integers. For all the propositions, let $a, b, c \in \mathbb{Z}$.

Proposition 7.1. *$a > b$ if and only if $a - b$ is a positive number.*

Proof. Suppose $a > b$. By definition of order, there exists some natural number k such that $a = b+k$ and $a \neq b$. Hence, by Lemma 1, $k \neq 0$, and k therefore is a positive number. In the other direction, suppose $a - b$ is a positive number. Define $k := a - b \neq 0$, so $a = b + k$, and hence $a > b$. By Lemma 1, since $k \neq 0$, we have that $a \neq b$. Therefore, $a > b$. □

Proposition 7.2. *Addition preserves order. In other words, if $a > b$, then $a + c > b + c$.*

Proof. Suppose $a > b$. Then, $a = b+k$ for some positive number k. Adding c to both sides of the equation gives $(a + c) = (b + c) + k$. Hence, $a + c > b + c$. □

Proposition 7.3. *Positive multiplication preserves order. If $a > b$ and c is positive, then $ac > bc$.*

Proof. Suppose $a > b$, so $a = b + k$ for some positive number k. Notice $ac = (b + k)c = bc + kc$. Since $c \neq 0$ and $k \neq 0$, we have $kc \neq 0$. Therefore, $ac > bc$. □

Proposition 7.4. *Negation reverses order. If $a > b$, then $-a < -b$ (i.e., $-b > -a$).*

Proof. Suppose $a > b$, so $a = b+k$ for some positive number k. We take the negation to get $-a = -b-k$. Notice that $-b = -a + k$, so $-b > -a$. □

Proposition 7.5. *Order is transitive. If $a > b$ and $b > c$, then $a > c$.*

Proof. Suppose $a > b$ and $b > c$, so $a = b + k$ and $b = c + l$ for positive numbers k and l. Notice, $a = b + k = c + l + k = c + (l + k)$. Since $l + k$ is also a positive number, we have $a > c$. □

Proposition 7.6. *Order is trichotomous. Exactly one of the statements hold $a > b, a < b$, or $a = b$ is true.*

Proof. Consider any two integers a and b. By the trichotomy of the integers, we have exactly one of these to be true: $a - b > 0, a - b < 0$, or $a - b = 0$. We consider each case and how it relates to the proposition.

1. Say $a - b > 0$, so $0 + k = (a - b)$ for some positive number k. By rearrangement, $b + k = a$, so $a > b$.

2. Say $a - b < 0$, so $(a - b) + l = 0$ for some positive number l. By rearrangement, $a + l = b$, so $a < b$.

3. Say $a - b = 0$. Then, $a = b$.

□

Problem 8 - Principle of Induction is only for the natural numbers

Show that the principle of induction does not apply directly to the integers. More precisely, given an example of a property $P(n)$ pertaining to an integer n such that $P(0)$ is true, and that $P(n)$ implies $P(n{+}{+})$ for all integers n, but that $P(n)$ is not true for all integers n. Thus, induction is not as useful a tool for dealing with the integers as it is with the natural numbers.

Proof. Consider the property $P(n) : n \geq 0$. Trivially, $P(0)$ is true. To show that the inductive hypothesis holds, suppose that $P(n)$ is true for any integer n, so $n \geq 0$. Clearly, $n{+}{+} \geq 0$, so $P(n{+}{+})$ holds true. However, $P(-1)$ is false.

The more interesting question though posed by Issa Rice is "What must the solution space look like?" More precisely, we can ask "Is there some simple characterization of every property P that works as a solution to this exercise?" The answer is that for any property P, its solution space will look the same as the set of all integers greater than some non-positive integer m. In other words, $\{n \in \mathbb{Z} : P(n)\} = \{n \in \mathbb{Z} : n \geq m\}$ for some $m \leq 0$. □

B: The Rationals

Definition 2 - Rationals ($\mathbb{Q}$)

A rational number is an expression of the form $a \,//\, b$, where a and b are integers and b is non-zero. Two rational numbers are considered equal ($a \,//\, b = c \,//\, d$) if and only if $ad = bc$.

Problem 1 - Equality on the rationals follow the three axioms

Verify that the definition of equality on the rationals is reflexive, symmetric, and transitive.

Proof. First, we show that the equality is reflexive. Consider the rational $a \,//\, b$. Notice that $ab = ba$. Hence, $a \,//\, b = a \,//\, b$.

Second, we prove that the equality is symmetric. Suppose $a \,//\, b = a' \,//\, b'$, so $ab' = a'b$. We can reverse the right and left hand sides to get $a'b = ab'$. Then, we use the commutativity of multiplication for the integers to get $a'b = b'a$, so $a' \,//\, b' = a \,//\, b$. □

Problem 2 - Addition, multiplication, and negation are well-defined

Let $a \,//\, b, c \,//\, d \in \mathbb{Q}$. We define addition as $a \,//\, b + c \,//\, d = (ad + bc) \,//\, (bd)$; multiplication as $(a \,//\, b) \times (c \,//\, d) = (ac) \,//\, (bd)$; and negation as $-(a \,//\, b) = (-a) \,//\, b$.

Proposition 2.1. *Addition on the rationals is well-defined.*

Proof. This has been done in the book. □

Proposition 2.2. *Multiplication on the rationals is well-defined.*

Proof. Suppose $a \,//\, b = a' \,//\, b'$, so $ab' = ba'$. We show that $(a \,//\, b) \times (c \,//\, d) = (a' \,//\, b') \times (c \,//\, d)$. By definition, the LHS is $(ac) \,//\, (bd)$, and the RHS is $(a'c) \,//\, (b'd)$. Notice:

$$\begin{aligned}
(a \,//\, b) \times (c \,//\, d) = (a' \,//\, b') \times (c \,//\, d) &\iff \big((ac) \,//\, (bd)\big) = \big((a'c) \,//\, (b'd)\big) && \text{by definition of multiplication} \\
&\iff (ac)(b'd) = (bd)(a'c) && \text{by definition of equality} \\
&\iff acb' = ba'c && \text{by cancellation law}
\end{aligned}$$

We now consider two cases: $c = 0$ or $c \neq 0$.

1. Say $c = 0$. Then $acb' = ba'c$ implies $0 = 0$.
2. Say instead that $c \neq 0$. Then, by the cancellation law, $ab' = ba'$. This is true by our supposition.

In either case, we see that the equality of $\big((ac) \,//\, (bd)\big) = \big((a'c) \,//\, (b'd)\big)$ holds because the reversible rearrangements lead to $ab' = ba'$, which is true by supposition. The proof holds similarly for the case of $c \,//\, d = c' \,//\, d'$.

□

Proposition 2.3. *Negation on the rationals is well-defined.*

Proof. Suppose $a \,//\, b = a' \,//\, b'$, so $ab' = ba'$. We show that $-(a \,//\, b) = -(a' \,//\, b')$. By definition of negation, the LHS is $(-a) \,//\, b$ while the RHS is $(-a') \,//\, b'$. Notice:

$$\begin{aligned}
-\big(a \,//\, b\big) = -\big(a' \,//\, b'\big) &\iff (-a) \,//\, b = (-a') \,//\, b' && \text{by definition of negation} \\
&\iff (-a)b' = b(-a') && \text{by definition of multiplication} \\
&\iff ab' = ba' && \text{by cancellation law}
\end{aligned}$$

In either case, we see that the equality of $-\big(a \,//b\big) = -\big(a' \,//b'\big)$ holds because the reversible rearrangements lead to $ab' = ba'$, which is true by supposition. □

Problem 3 - Laws of Algebra for the Rationals

Let $x = x_1 \,//\, x_2, y = y_1 \,//\, y_2, z = z_1 \,//\, z_2$ be rationals where $x_1, x_2, y_1, y_2, z_1, z_2 \in \mathbb{Z}$. Then, the following laws of algebra hold.

Proposition 3.1. $x + y = y + x$

Proof. Notice:

$$\begin{aligned} x + y &= x_1 \,//\, x_2 + y_1 \,//\, y_2 \\ &= (x_1y_2 + x_2y_1) \,//\, (x_2y_2) \\ &= (y_2x_1 + y_1x_2) \,//\, (y_2x_2) \\ &= (y_1x_2 + y_2x_1) \,//\, (y_2x_2) \\ &= y + x \end{aligned}$$

□

Proposition 3.2. $(x + y) + z = x + (y + z)$

Proof. This has been done in the book. □

Proposition 3.3. $x + 0 = 0 + x = x$

Proof. Notice:

$$\begin{aligned} x + 0 &= (x_1 \,//\, x_2) + (0 \,//\, 1) \\ &= (x_11 + x_20) \,//\, (x_21) \\ &= (1x_1 + 0x_2) \,//\, (1x_2) \\ &= x_1 \,//\, x_2 \\ &= x \end{aligned}$$

Because addition is commutative, we have that $0 + x = x$. □

Proposition 3.4. $x + (-x) = (-x) + x = 0$

Proof. Notice:

$$\begin{aligned} x + (-x) &= (x_1 \,//\, x_2) + (-(x_1 \,//\, x_2)) \\ &= (x_1 \,//\, x_2) + ((-x_1) \,//\, x_2) \\ &= (x_1x_2 + x_2(-x_1)) \,//\, (x_2x_2) \\ &= (x_1x_2 - x_1x_2) \,//\, (x_2x_2) \\ &= 0 \,//\, (x_2x_2) \\ &= 0 \end{aligned}$$

Because addition is commutative, we have that $(-x) + x = 0$. □

Proposition 3.5. $xy = yx$

Proof. Notice:

$$\begin{aligned} xy &= (x_1 \,//\, x_2) \times (y_1 \,//\, y_2) \\ &= (x_1y_1) \,//\, (x_2y_2) \\ &= (y_1x_1) \,//\, (y_2x_2) \\ &= yx \end{aligned}$$

□

Proposition 3.6. $(xy)z = x(yz)$.

Proof. Notice:

$$\begin{aligned} (xy)z &= \big((x_1 \,//\, x_2) \times (y_1 \,//\, y_2)\big) \times (z_1 \,//\, z_2) \\ &= \big((x_1y_1) \,//\, (x_2y_2)\big) \times (z_1 \,//\, z_2) \\ &= \big((x_1y_1)z_1\big) \,//\, \big((x_2y_2)z_2\big) \\ &= \big(x_1(y_1z_1) \,//\, x_2(y_2z_2)\big) \\ &= (x_1 \,//\, x_2) \times \big((y_1z_1) \,//\, (y_2z_2)\big) \\ &= (x_1 \,//\, x_2) \times \big((y_1 \,//\, y_2) \times (z_1 \,//\, z_2)\big) \\ &= x(yz) \end{aligned}$$

□

Proposition 3.7. $x1 = 1x = x$

Proof. Notice:

$$\begin{aligned} x1 &= (x_1 \,//\, x_2) \times (1 \,//\, 1) \\ &= (x_11) \,//\, (x_21) \\ &= (1x_1) \,//\, (1x_2) \\ &= (x_1) \,//\, x_2 \\ &= x \end{aligned}$$

Because multiplication is commutative, $1x = x$. □

Proposition 3.8. $x(y + z) = xy + xz$

Proof. There are diminishing returns if I do this. □

Proposition 3.9. $(y + z)x = yx + zx$

Proof. Notice:

$$\begin{aligned} (y + z)x &= x(y + z) \\ &= xy + xz \\ &= yx + zx. \end{aligned}$$

□

Proposition 3.10. *If x is non-zero, we have $xx^{-1} = x^{-1}x = 1$*

Proof. Suppose x is non-zero, so $x = a \,//\, b$ where a and b are non-zero integers. Notice:

$$\begin{aligned} xx^{-1} &= (x_1 \,//\, x_2) \times (x_1 \times x_2)^{-1} \\ &= (x_1 \,//\, x_2) \times (x_2 \times x_1) \\ &= (x_1x_2) \,//\, (x_2x_1) \\ &= (x_1x_2) \,//\, (x_1x_2) \\ &= 1 \end{aligned}$$

□

Problem 4 - Trichotomy of $\mathbb{Q}$

Let $x \in \mathbb{Q}$. We show the trichotomy of $\mathbb{Q}$, where exactly one of the following statements hold: $x = 0$, x is a positive rational number, or x is a negative rational number.

Proof. Let $x = a \,//\, b \in Q$ where a is is an integer and b is a non-zero integer. We consider six cases, which are based on the possible combinations of a and b being a positive integer, negative integer, or zero.

1. Say $a < 0$ and $b < 0$, so there exist positive natural numbers a', b' such that $a = -a'$ and $b = -b'$. By substitution, $a \,//\, b = (-a') \,//\, (-b')$. Notice also that $(-a') \,//\, (-b') = a' \,//\, b'$ because $(-a')b' = (-b')a'$. Hence, $a \,//\, b = a' \,//\, b'$ where the RHS is positive because $a', b' > 0$. Therefore, $a \,//\, b$ is positive.

2. Say $a < 0$ while $b > 0$, so there exists a positive natural number a' such that $a = -a'$. Hence, $a \,//\, b = (-a') \,//\, b = -(a' \,//\, b)$. Because $a \,//\, b$ is the negation of $a' \,//\, b$, $a \,//\, b$ is negative.

3. Say $a > 0$ and $b < 0$, so there exists a positive natural number b' such that $b = -b'$. By substitution, $a \,//\, b = a \,//\, (-b')$. Notice that because $ab' = (-b')(-a)$. we have that $a \,//\, (-b') = (-a) \,//\, b'$ by definition of equality. Observe that $(-a) \,//\, b' = -(a \,//\, b')$. Therefore, we have a long list of equalities as summarized:
$$a \,//\, b = a \,//\, (-b') = (-a) \,//\, b' = -(a \,//\, b')$$
This is all to just say that $a \,//\, b$ is the negation of the positive rational $a \,//\, b'$. Hence, $a \,//\, b$ is negative.

4. Say $a > 0$ and $b > 0$. By the definition of a positive rational, $a \,//\, b$ is also positive.

5. Say $a = 0$. Notice that for any $b \in \mathbb{N} \setminus \{0\}$, we have that $a \,//\, b = 0 \,//\, b = 0$ We show this through the equivalence eqaution. Notice:
$$\begin{aligned} 0 \,//\, b = 0 \,//\, 1 &\iff 0 \times 1 = b \times 0 && \text{by definition of equality} \\ &\iff 0 = 0 && \text{by multiplication} \end{aligned}$$

We have shown that for every rational, they are at least either $x = 0$, $x > 0$ or $x < 0$. However, we now need to show that this is mutually exclusive, so they are also AT MOST $x = 0$, $x > 0$ or $x < 0$. Suppose $x = 0$, so $x = 0 \,//\, b$. Hence, $a = 0$, so a is not a positive number. Therefore, x cannot be positive. Similarly, suppose for the sake of contradiction that $x = 0$ and $x < 0$. Hence, $x = 0 \,//\, 1$. Also, $x = (-c) \,//\, d$ for some positive numbers c, d. Then, $0 \,//\, 1 = (-c) \times d$, so $0d = (-c)1 = -c$. This implies that $c = 0$, but c is positive. A contradiction!

Suppose for the sake of contradiction that $x < 0$ and $x > 0$, so $x = a \,//\, b$ and $x = (-c) \,//\, d$ for positive numbers a, b, c, d. Then, $a \,//\, b = (-c) \,//\, d$ implies $ad = b(-c) = -(bc)$. Since a, b, c, d are positive, ad and bc are positive, and $-(bc)$ is negative. However, since $ad = -(bc)$, we now have that ad is also negative. By the trichotomy of integers, ad can only be at most positive, negative, or zero. A contradiction!

□

These three lemmas are here for my sanity because I really want to show that if you multiply by a positive number, the sign does not change. However, if you multiply by a negative number, the sign does flip. I just add these lemmas for completeness and especially because I will be using them for the next problem.

Lemma 2. *Let $x = x_1 \,//\, x_2$ and $y = y_1 \,//\, y_2$ be rational numbers, where x_1, x_2, y_1, y_2 are integers. If y is positive, then xy has the same sign as x.*

Proof. Suppose y is positive, so y_1 and y_2 are positive integers. We break the proof down into three cases of x being positive, zero, or negative.

1. Say x is positive, so x_1 and x_2 are also positive. Then, $xy = (x_1y_1) \,//\, (x_2y_2)$ is also positive because x_1y_1 and x_2y_2 are positive.

2. Say x is zero. Then, $xy = 0y = 0$.

3. Say x is negative, so there exists a positive rational number x' such that $x = -x'$. Then, $xy = (-x')y = -(x'y)$. Since x' and y are positive, it follows from Case (1) that $x'y$ is positive. Hence, xy is the negation of the positive rational $x'y$. Therefore, xy is negative.

In the three cases, we have shown that xy has the same sign as x. □

Lemma 3. *Multiplying by -1 is the same as negation for the rationals too (i.e., $\forall x \in \mathbb{Q}$, $(-1) \times x = -x$.)*

Proof. Let $x = x_1 \mathbin{//} x_2 \in \mathbb{Q}$ where $x_1, x_2 \in \mathbb{Z}$. Notice:

$$\begin{aligned}
(-1) \times x &= (-1 \mathbin{//} 1) \times (x_1 \mathbin{//} x_2) && \text{by substitution of } -1 \text{ and } x \\
&= (-1x_1) \mathbin{//} (1x_2) && \text{by definition of multiplication on the rationals} \\
&= (-x_1) \mathbin{//} (x_2) && \text{by definition of multiplication on the integers} \\
&= -(x_1 \mathbin{//} x_2) && \text{by definition of negation} \\
&= -x && \text{by substitution}
\end{aligned}$$

□

Lemma 4. *The negation of a negative rational number $-(-x)$ is positive.*

Proof. Let $-x$ be a negative rational number. Observe that $-(-x) = (-1) \times \big((-1) \times x\big) = (-1 \times -1) \times x = 1 \times x = x$. □

Problem 5 - Properties of the order on the rationals

For the properties of the order on the rationals, let $x, y, z \in \mathbb{Q}$. Then, the following properties hold.

Proposition 5.1. *(Order Trichotomy) Exactly one of the three statements hold: (i) $x = y$; (ii) $x < y$; (iii) $x > y$.*

Proof. Consider the rational $x - y$. By the law of trichotomy for rationals, $x - y$ can at most be negative, zero, or positive. If $x - y$ is negative, then $x < y$. If $x - y = 0$, then $x = y$. Finally, if $x - y$ is positive, then $x > y$. This is mutually exclusive and collectively exhaustive by the law of trichotomy. Hence, exactly one of the statements hold for any rational numbers x and y. □

Proposition 5.2. *(Order is anti-symmetric) We have that $x < y$ if and only $y > x$.*

Proof. Suppose $x < y$, so $x - y$ is negative. Notice that the negation $-(x - y) = y - x$ is positive by Lemma 4. Hence, $y > x$.

Conversely, suppose that $y > x$, so $y - x$ is negative. Notice that the negation $-(y - x) = x - y$ is positive by Lemma 4. Hence, $x > y$. □

Proposition 5.3. *(Order is transitive) If $x < y$ and $y < z$, then $x < z$.*

Proof. Suppose $x < y$ and $y < z$, so $x - y$ and $y - z$ is negative. Therefore, there exist positive rational numbers u and v such that $x - y = -u$ and $y - z = -v$. If we add $-u$ to $-v$, we get $(-u) + (-v) = (-1 \times u) + (-1 \times v)$ by Lemma 3. We then use the distributive property to have that $(-1 \times u) + (-1 \times v) = -1(u + v)$. Overall, we have that $(-u) + (-v) = -(u + v)$. Since $u + v$ is a positive number, we have that $(-u) + (-v)$ is a negative number since it is the negation of $u + v$.

Notice also that $(-u) + (-v) = (x - y + y - z) = x - z$. Overall, we now know that $x - z$ is a negative number. Therefore, $x < z$. □

Proposition 5.4. *(Addition preserves order) If $x < y$, then $x + z < y + z$.*

Proof. Suppose $x < y$, so $x - y$ is a negative rational number. Notice that $(x + z) - (y + z) = x + z - y - z = x - y$ is also a negative rational number. Therefore, $x + z < y + z$. □

Proposition 5.5. *(Positive multiplication preserves order) If $x < y$ and z is positive, then $xz < yz$.*

Proof. Suppose $x < y$ and z is postive. Then, so $x - y$ is negative. By Lemma 2, $(x - y)z = xz - yz$ is still negative. Hence, $xz < yz$. □

Problem 6 - Multiplying by a negative number reverses order.

Show that if x, y, z are rational numbers such that $x < y$ and z is negative, then $xz > yz$.

Proof. Let x, y, z be rational numbers. Suppose $x < y$ and z is negative. By the first supposition, $x - y$ is negative, so there exists a positive rational u such that $x - y = -u$. By the second supposition, there exists a positive rational number v such that $z = -v$. Notice that $(-u)(-v) = (-1 \times u)(-1 \times v)$ by Lemma 3. Then, we can use the commutativity and associativity of multiplication to get $(-u)(-v) = (-1 \times -1)(uv) = 1(uv) = uv$. This is a very long process to just say $(-u)(-v) = uv$. By Lemma 2, since u and v are both positive, uv is positive. Hence, $(-u)(-v)$, which is equal to uv, is also positive.

Now, observe also that $(-u)(-v) = (x - y)z = xz - yz$. Since we have already previously determined that $(-u)(-v)$ is positive, it follows that $xz - yz$ is positive. Therefore, $xz > yz$. □

C: Absolute Value and Exponentials

Definition 3 - Absolute Value $|\cdot|$

If x is a rational number, the absolute value of $|x|$ if x is defined as follows: If x is negative, then $|x| := -x$. If x is zero, then $|x| := 0$. If x is positive, then $|x| := x$.

Definition 4 - Distance $d(\cdot,\cdot)$

Let x and y be rational numbers. The quantity $|x-y|$ is called the distance between x and y and is sometimes denoted as $d(x,y) := |x-y|$.

Definition 5 - Exponentiation to an integer

Let $x \in \mathbb{Q}$. To raise x to the power 0, we define $x^0 := 1$; in particular, we define $0^0 := 1$. Now, suppose inductively that x^n has been defined for some $n \in \mathbb{N}$. Then, we define $x^{n+1} := x^n \times x$. If instead we wish to raise x to the power of a negative number, we need that x is a non-zero rational and have a negative integer $-n$. We define $x^{-n} := \frac{1}{x^n}$.

Lemma 5. *We have that $|x|^2 = x^2$.*

Proof. Let x to be a rational number. We consider three cases by the law of trichotomy on the rationals.

1. Say x is negative, so $|x|^2 = (-x)^2 = x^2$.
2. Say $x = 0$. Observe $|x|^2 = |0|^2 = 0^2$.
3. Say x is positive. Then, $|x|^2 = x^2$.

In all of the three cases, we see that $|x|^2 = x^2$. ☐

Problem 1 - Properties of absolute value and distance

Verify the basic properties of absolute value and distance. Let $x, y, z \in \mathbb{Q}$.

Proposition 1.1. *Non-degeneracy of absolute value. We have $|x| \geq 0$. Also, $|x| = 0$ if and only if $x = 0$.*

Proof. First, we show that $|x| \geq 0$. By the law of trichotomy, we consider three cases. If x is negative, then there exists a positive rational number x' such that $x = -(x')$. Then, $|x| = |-(x')| = x'$ (by multiplicativity of absolute value). Since x' is positive, it follows that $|x| \geq 0$. If $x = 0$, then clearly $|x| = |0| = 0 \geq 0$. Finally, if x is positive, then $|x| = x \geq 0$.

Now, we show that $|x| = 0$ if and only if $x = 0$. Now, suppose that $x \neq 0$. We now consider two cases. If x is negative, then $|x| = -x$ is positive. By the law of trichotomy, $|x|$ cannot be 0. If x is instead positive, then $|x =| = x$ is also positive, which by the law of trichotomy, cannot be 0. Hence, in either case, $|x| \neq 0$. Conversely, suppose that $x = 0$. Trivially, $|x| = |0| = 0$. ☐

Proposition 1.2. *(Triangle inequality for absolute value) We have $|x+y| \leq |x| + |y|$.*

Proof. We wish to show $|x+y| \leq |x| + |y|$. By definition of order on the rationals, we need to show that

$$|x+y| - (|x| + |y|) \leq 0.$$

To prove this, we now consider two cases of $x + y = 0$ and $x + y \neq 0$.

1. Say $x + y = 0$. Then, $|x+y| - (|x| + |y|) = 0 - (|x| + |y|) = -(|x| + |y|) \leq 0$. Therefore, $|x+y| - (|x| + |y|) \leq 0$.

2. Say instead that $x + y \neq 0$, so $x \neq 0$ and $y \neq 0$. Then, we will use Lemma 5 whenever we have $|x|^2 = x^2$. Notice:

$$\begin{aligned}
|x+y| - (|x|+|y|) \leq 0 &\iff \big(|x+y| - (|x|+|y|)\big)\big(|x+y| + (|x|+|y|)\big) \leq 0 \\
&\iff |x+y|^2 - \big(|x|+|y|\big)^2 \leq 0 \\
&\iff (x+y)^2 - (|x|^2 + 2|x||y| + |y|^2) \leq 0 \\
&\iff x^2 + 2xy + y^2 - (x^2 + 2|x||y| + y^2) \leq 0 \\
&\iff 2xy - 2|x||y| \leq 0 \\
&\iff xy - |x||y| \leq 0
\end{aligned}$$

We will now just use cases to show that $xy - |x||y| \leq 0$. (We can also be smarter by using work that we will do because Propositions 1.3 and 1.4 give us the tools to skip the casework.)

(a) Say $x < 0$ and $y < 0$, so there exist positive rational numbers x' and y' such that $x = -(x')$ and $y = -(y')$. Then $xy - |x||y| = (-x')(-y') - |-x'||-y'| = x'y' - x'y' = 0 \leq 0$.

(b) Say $x < 0$ and $y > 0$, so there exists a positive rational number x' such that $x = (-x')$. Then, $xy - |x||y| = (-x')y - |-x'||y| = -x'y - x'y = -2x'y = -2(x'y) < 0 \leq 0$

(c) Say $x > 0$ and $y < 0$. The logic is the same as for the proof above because of symmetry between the x and y. In any case, we will have that $xy - |x||y| < 0 \leq 0$.

(d) Say $x > 0$ and $y > 0$, so $xy - |x||y| = xy - xy = 0 \leq 0$.

In all of the four cases, we have that $xy - |x||y| \leq 0$. Therefore, we can say that $|x+y| - (|x|+|y|) \leq 0$.

Overall, in either case of $x + y = 0$ or $x + y \neq 0$, we have shown that $|x+y| \leq (|x| + |y|)$ ☐

Proposition 1.3. *We have the inequalities $-y \leq x \leq y$ if and only if $y \geq |x|$. In particular, we have $-|x| \leq x \leq |x|$.*

Proof. s ☐

Proposition 1.4. *(Multiplicativity of absolute value) We have $|xy| = |x||y|$. In particular, $|-x| = |x|$.*

Proof. s ☐

Proposition 1.5. *(Non-degeneracy of distance) We have $d(x, y) \geq 0$. Also, $d(x, y) = 0$ if and only if $x = y$.*

Proof. smth ☐

Proposition 1.6. *(Symmetry of distance) We have $d(x, y) = d(y, x)$.*

Proof. sh ☐

Proposition 1.7. *(Triangle inequality for distance) We have $d(x, z) \leq d(x, y) + d(y, z)$.*

Proof. chu ☐

Definition 6 - ε-closeness

Let $\varepsilon > 0$ be a rational number, and let x, y be rational numbers. We say that y is ε-close to x if and only if we have that $d(x, y) \leq \varepsilon$.

Problem 2 - Properties of ε-closeness

Let $x, y, z, w \in \mathbb{Q}$. Then, the following properties hold:

Proposition 2.1. *The number x is ε-close to y for every $\varepsilon > 0$ if and only if $x = y$.*

Proof. d ☐

Proposition 2.2. *Let $\varepsilon > 0$. If x is ε-close to y, then y is ε-close to x.*

Proof. s ☐

Proposition 2.3. *Let $\varepsilon, \delta > 0$. If x is ε-close to y, and y is δ-close to z, then x and z are $(\varepsilon + \delta)$-close.*

Proof. chuchu □

Proposition 2.4. *Let $\varepsilon > 0$. If x and y are ε-close, they are also ε'-close for every $\varepsilon' > \varepsilon$.*

Proof. chuchu □

Proposition 2.5. *Let $\varepsilon > 0$. If y and z are both ε-close to x and w is between y and z (i.e., $y \leq w \leq z$ or $z \leq w \leq y$), then w is also ε-close to x.*

Proof. chuchu □

Proposition 2.6. *Let $\varepsilon > 0$. If x and y are ε-close, and z is non-zero, then xz and yz are $\varepsilon|z|$-close.*

Proof. □

Proposition 2.7. *Let $\varepsilon, \delta > 0$. Suppose x and y are ε-close, and z and w are δ-close. Then xz and yw are $(\varepsilon|z| + \delta|x| + \varepsilon\delta)$-close.*

Problem 3 - Properties of Exponentiation I

Let x, y be rational numbers. Let n, m be natural numbers. The following properties hold.

Proposition 3.1. *We have $x^n x^m = x^{n+m}$.*

Proof. chuchu □

Proposition 3.2. *We have $(x^n)^m = x^{nm}$.*

Proof. chuchu □

Proposition 3.3. $(xy)^n = x^n y^n$.

Proof. chuchu □

Proposition 3.4. *Suppose $n \geq 0$. Then, we have $x^n = 0$ if and only if $x = 0$.*

Proof. suppose □

Proposition 3.5. *If $x \geq y \geq 0$, then $x^n \geq y^n \geq 0$.*

Proof. suppose □

Proposition 3.6. *If $x > y \geq 0$ and $n > 0$, then $x^n > y^n \geq 0$.*

Proof. suppose □

Proposition 3.7. *We have $|x^n| = |x|^n$.*

Proof. suppose □

Problem 4 - Properties of Exponentiation II

Let x, y be rational numbers. Let n, m be integers. The following properties hold.

Proposition 4.1. *We have $x^n x^m = x^{n+m}$.*

Proof. chuchu □

Proposition 4.2. *We have $(x^n)^m = x^{nm}$.*

Proof. chuchu □

Proposition 4.3. $(xy)^n = x^n y^n$.

Proof. chuchu □

Proposition 4.4. *If $x \geq y > 0$, then $x^n \geq y^n > 0$ if n is positive OR $0 < x^n \leq y^n$ if n is negative.*

Proof. suppose □

Proposition 4.5. *If $x, y > 0$, $n \neq 0$, and $x^n = y^n$, then $x = y$.*

Proof. suppose □

Proposition 4.6. *We have $|x^n| = |x|^n$.*

Proof. suppose □

Problem 5 - Exponentiation grows faster than linear

Prove that $2^N \geq N$ for all positive integers N.

Proof. chuchu □

D: Gaps in the Rational Numbers

Problem 1 - Interspesing of integers by rationals

Let x be a rational number. Then, there exists an integer n such that $n \leq x < n+1$. In fact, this integer is unique (i.e., for each x, there is only one n for which $n \leq x < n+1$). In particular, there exists a natural number N such that $N > x$ (i.e., there is no succh thing as a rational number which is larger than all the natural numbers.)

Proof. □